

Emily Choong

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Skills

Tools: Python, SQL, Excel, Tableau, Amazon Bedrock, R, C/C++, PowerPoint, Claude

Skills: Statistical Modelling, Machine Learning (XGBoost, Random Forest), NLP, Data Classification & Normalisation, AI agents

Education

University of Bristol

Bristol, England

BSc Data Science with a Year in Industry: On track for 1st class honours

2023-2027

Relevant Units: Advanced Linear Modelling and Classification, Probability & Statistics, Algorithms and Machine Learning, Financial Mathematics, Time Series and Stochastic Processes

Shrewsbury School

Shrewsbury, England

A*A*A*A*: Math, Further Math, Chemistry, Physics

2022-2024

Work Experience & Projects

Data Science Intern (Year in Industry)

Sep 2025 - 2026

Pacific Life Re, London

Survival Risk Modelling from UK Primary Care Data

- Built and cleaned a large-scale patient dataset (~2M individuals, 30+ tables) from linked EHR sources, developing a reproducible Python pipeline on AWS (SageMaker, Redshift) to resolve missingness, inconsistencies, and schema alignment for a modelling-ready cohort.
- Engineered predictive features from clinical data (e.g. blood pressure, cholesterol, BMI, lifestyle), applying research-backed banding, handling ~50% missingness via MICE with sensitivity analysis to improve robustness and interpretability.
- Developed and evaluated survival models (Cox PH via lifelines and XGBoost survival) to analyse drivers of time-to-event outcomes, validating assumptions and separating true effects from confounding.

Cause of Claim Classifier - using NLP

- Built an NLP pipeline using TF-IDF and a hierarchical Random Forest classifier to categorise free-text insurance claims into a three-level taxonomy of 256 labels, achieving 95% accuracy and automating classification for over 90% of records
- Deployed the model across 20,000+ claim reasons within the team's data pipeline, directly informing R&D actuarial assumptions and insurer performance analysis
- Led the integration of LLMs into the pipeline to improve scalability and efficiency, collaborating with risk and compliance teams to enable safe adoption and drive AI-powered capabilities into core business workflows.

Data Schema AI agent

- Built an AI agent to help new users understand and navigate a complex database, improving accessibility and reducing onboarding time.
- Used Python to scrape and core tables across 5 schemas (~2 billion rows), generating a structured JSON knowledge base detailing fields, data ranges, examples, and join keys, creating the agent's knowledge base.
- Gathered feedback from users, improving precision and readability in responses by applying prompt engineering and custom instructions.

UW Normalisation (structuring underwriting data for risk & pricing)

- Transform 500,000+ unstructured underwriting Q&A pairs into 9 interpretable risk markers using text embeddings and K-means clustering, standardising a dataset of 2.2M records
- Build a decision-tree model (held-out AUC 0.977) showing the markers cleanly separated declined from standard risks, and quantify where they did and did not predict pricing loadings, directly shaping how the markers were used in downstream pricing and experience analysis
- Communicated findings effectively to non-technical underwriting and medical actuarial teams, leading with plain-language statistics (e.g. 91% of declined customers flagged "non-compliant with treatment" vs 2% of standard) to build commercial buy-in

Product Intern - AI Biotech startup

Jul - Aug 2025

Dr KuDos Lab, Hong Kong

- Redesigned the onboarding flow to shorten the path to a new user's first meaningful action which lowered onboarding friction increasing customer acquisition
- Identified features to drive ongoing engagement for higher retention including a gamified points system and a patient diary; the diary was approved by the founder for the product roadmap

Intern - Private Equity Firm

Aug - Sep 2024

Triple P Capital, Singapore

- Researched investment risks and opportunities in the Philippine bancassurance sector, streamlined findings into concise formats, supporting analysis and decision-making
- Designed the firm's website end to end, owning visual layout, structure and user experience to present the firm clearly to prospective clients

Extracurriculars & Interests

Badminton Club Lead – Pacific Life Re; organised regular matches and monthly social events for cross-team bonding.

Captain of Badminton team – Dunman High School; Build communication and team spirit within the team.

Secretary of Malaysian Badminton Society – University of Bristol; Organise weekly social training sessions with 18 attendees per session.

Languages: English (Native), Mandarin (Proficient)

Music: Grade 8 in Piano, Violin & Music Theory

Interests: Yoga, Badminton, Running – participated in the Bath Half Marathon (2025), Reading